Northern Illinois University

CSCI 490-B9 – Deep Learning

Course Project, Part 2: The Progress Report

Fall 2026

Released:	Tuesday, November 10, 2026
Suggested date:	Thursday, November 19, 2026, 11:59pm
Latest date:	Thursday, November 26, 2026, 11:59pm
Requires:	a proposal with at least 4 of 5 points
Weight:	5% of your course grade
Submit:	on Blackboard: the link to one commit of your GitHub repository (Part 1, Appendix A.6)

Milestones on your own timeline

Everyone moves at a different pace, so the project milestones now have two dates instead of one deadline. **This replaces the due dates and the late policy in Part 1.** The presentation and the final report keep fixed dates, because they cannot move.

Milestone	Shows	Suggested	Latest	Needed before
Proposal (5%)	Level 0, data check, plan	Thu 11/5	Tue 11/17	the progress report
Progress report (5%)	Level 1	Thu 11/19	Thu 11/26	the presentation
Presentation (10%)	Levels 0–2	in class, Tue 12/1 or Thu 12/3 (fixed)		the final report
Final report (15%)	all levels, Level 3	finals week, date on Blackboard (fixed)		–

Each milestone opens the next. A milestone is graded only after the one before it has at least **80%** (4 of 5 points for the proposal and the progress report). The reason is simple: Level 1 is compared with Level 0, and Level 2 with Level 1. A progress report without a working Level 0 has nothing to compare with.

Earlier is worth more. You may submit any time before the latest date, with no penalty per day. What changes is what you get back:

You submit	You get
by the suggested date	written feedback (an issue in your repository) and **one revision**: fix what the feedback lists, post the new commit link in the issue within 5 days, and get the points back
after the suggested date, by the latest date	a grade, once; no revision for points
after the latest date	no points, but the milestone still has to reach 80% to open the next one. It is never too late to do a milestone, only too late to score it

If a milestone you submitted late gets less than 80%, fix it and post the new commit link: the fix opens the next milestone, and the grade stays as it was.

No proposal yet? You do Task A unless your own task was approved. Follow Part 1 and submit it by Tuesday 11/17; the progress report is graded once your proposal reaches 4 of 5 points. An example of a full-marks Task A proposal is on Blackboard (`proposal_example_task_a.md`). Read it for the level of detail, then write your own.

One minute on Tuesdays. Each Tuesday class starts with a round: which level you are on, and what is in your way. It is not graded. Its purpose is that nobody is stuck alone for a week.

The progress report: Level 1

Level 1 is *a neural model you train yourself*: you write or adapt the model, you train it from scratch or from word vectors, and you choose its settings on the validation set. A pretrained Transformer is Level 2, not Level 1.

Task	Level 1 models to start from
A (venue)	the embedding bag or the 1D CNN from the 11/10 lecture; an MLP on TF-IDF features; a BiLSTM
B (citation)	the same models on the context. Keep `[CITATION]` as one token, consider adding the section name, and weight the core class (18%); choose the threshold on validation
C (links)	a GNN for link prediction, for example GraphSAGE on the known edges, with TF-IDF features reduced by SVD as node features. Hide part of the known edges, learn to recover them, and rank the candidates with `evaluate_c`
own task	a neural model you train, in the form your proposal promised

A Level 1 model that ties Level 0 is a good result if the comparison is fair and you can say why. On Task A, the embedding bag and the CNN from the lecture tie TF-IDF (about 42% on validation); that is a finding, not a failure.

What to commit

1. **A script per model**, for example `level1_bag.py`, that loads the data with `mustcite.py`, trains, and saves its metrics with `save_results()`. It fixes its random seed and runs in under 30 minutes on a free Colab T4 (`run_on_colab.ipynb` shows how to run it there).

2. **Three seeds.** Train your final setting with seeds 0, 1 and 2, and report the mean ± the standard deviation. Differences between models on these tasks are often one or two points; with one seed you cannot tell a better model from a lucky run.

3. **Choices on validation, test once.** Pick the number of epochs, the learning rate and anything else on the validation set. Then evaluate the final setting on the test set once per seed.

4. **A results file per run**: `results/level1_<model>_seed<k>_val.json` and `_test.json`. Every number in the README comes from these files.

5. **One figure**: training and validation loss (or metric) per epoch for one seed, saved in `figures/`. It shows where you stopped and whether the model overfits.

6. **The README**, updated as below.

The README: what changes

Keep sections 1–6 from the proposal (fix them if the feedback asked you to). Then:

1. **Results table.** One row per level, with the files it comes from:

Level	Model	Validation	Test	Results files
0	TF-IDF + logistic regression	42.4	39.2	`level0_task_a_*.json`
1	Embedding bag, $d = 64$	42.1 ± 0.3	39.9 ± 0.3	`level1_bag_seed{0,1,2}_*.json`

2. **A new section "7. Progress report"** with four short parts:

   (a) *What I built.* The model in two or three sentences: its input, its layers, the number of parameters, and the settings you chose on validation, with the values you tried.

   (b) *How it compares.* Against Level 0, and against the number you expected in the proposal. Look past the average: for Task A, the confusion matrix or per-venue recall; for Task B, precision and recall; for Task C, which queries it gets right. Include your figure.

   (c) *What works, what does not.* What you tried that did not help counts too; say what you tried and what happened.

   (d) *What I will change.* Your Level 2 model, and your Level 3 question, revised if Level 1 taught you something.

3. **The AI tool note** at the top, updated for this milestone.

Grading of the progress report (5% of the course grade)

I grade the commit you link to, not the later state of your repository.

Part	Share	Full marks when
Level 1 model and code	25%	a neural model you train; the script runs from the repository, fixes its seed and saves results
Results	30%	three seeds as mean ± sd; choices made on validation, test once; README numbers match `results/`
Comparison	25%	against Level 0 and the expected number, with a look past the average and the figure
What works and what next	10%	concrete: what was tried, what happened, what changes
Updated plan	10%	the Level 2 model and a concrete Level 3 question

Submission checklist

1. Your proposal has at least 4 of 5 points (or you posted its fix).
2. `level1_<model>.py` is committed, and running it reproduces your results files.
3. `results/` has a validation file and a test file for each of the three seeds.
4. The README's results table matches the files; section 7 has all four parts; the figure is committed.
5. Everything is pushed, and the commit you link has all of it: open the link in a private browser window and check before you post.
6. On Blackboard, paste the link to that commit.

Appendix: three seeds in practice

Fix every source of randomness at the start of each run, keep the best epoch on validation, and save one file per seed and split:

```
import random, numpy as np, torch
from mustcite import evaluate_a, save_results

for seed in (0, 1, 2):
    random.seed(seed); np.random.seed(seed); torch.manual_seed(seed)
    model = Bag(V).to(device)
    best_epoch, best_state = train(model, train_loader, val_loader) # on validation
    model.load_state_dict(best_state)
    config = {"model":  "bag", "d":  64, "seed":  seed, "best_epoch":  best_epoch}
    for split, df, loader in (("val", val, val_loader), ("test", test, test_loader)):
        metrics = evaluate_a(df.label, predict(model, loader))
        save_results(f"level1_bag_seed{seed}_{split}", metrics, config)
```

Then read the six files back and print the mean and the standard deviation for the README table. Ask your agent for that script too; check its numbers against the files by hand once.